

DIgSILENT Technical Documentation

Short-Circuit Calculation according to IEC 60909

Comparison Report



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Comparison Report

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1 Overview

The following report compares the IEC short circuit results obtained with DIgSILENT PowerFactory software[1] with the results provided in the Technical Report 60909-4 [1]. In this document only the test cases **Example 1**(Low Voltage System) and **Example 4** (High Voltage Test Network) are compared for both three-phase and line to earth short circuit currents. Suggestions on the entry of equipment data as well as the options in the short circuit command window in *PowerFactory* are also detailed.

[1] Version 14.1.3

2 Example 1: IEC Low Voltage System

The following example is taken from the IEC Technical Report TR 60909-4[1], clause 3, page 37. In this example, a low voltage system of 400 Volts with a frequency of 50 Hz is given, in which, the short circuit currents I''_k and i_p are calculated at locations F1, F2 and F3.

2.1 System Description

The low voltage system is illustrated in Figure 2.1. The system depicts a 20 kV network feeder connected to a 400 V system via two step-down transformers. The short circuit currents are calculated at F1, F2 and F3.

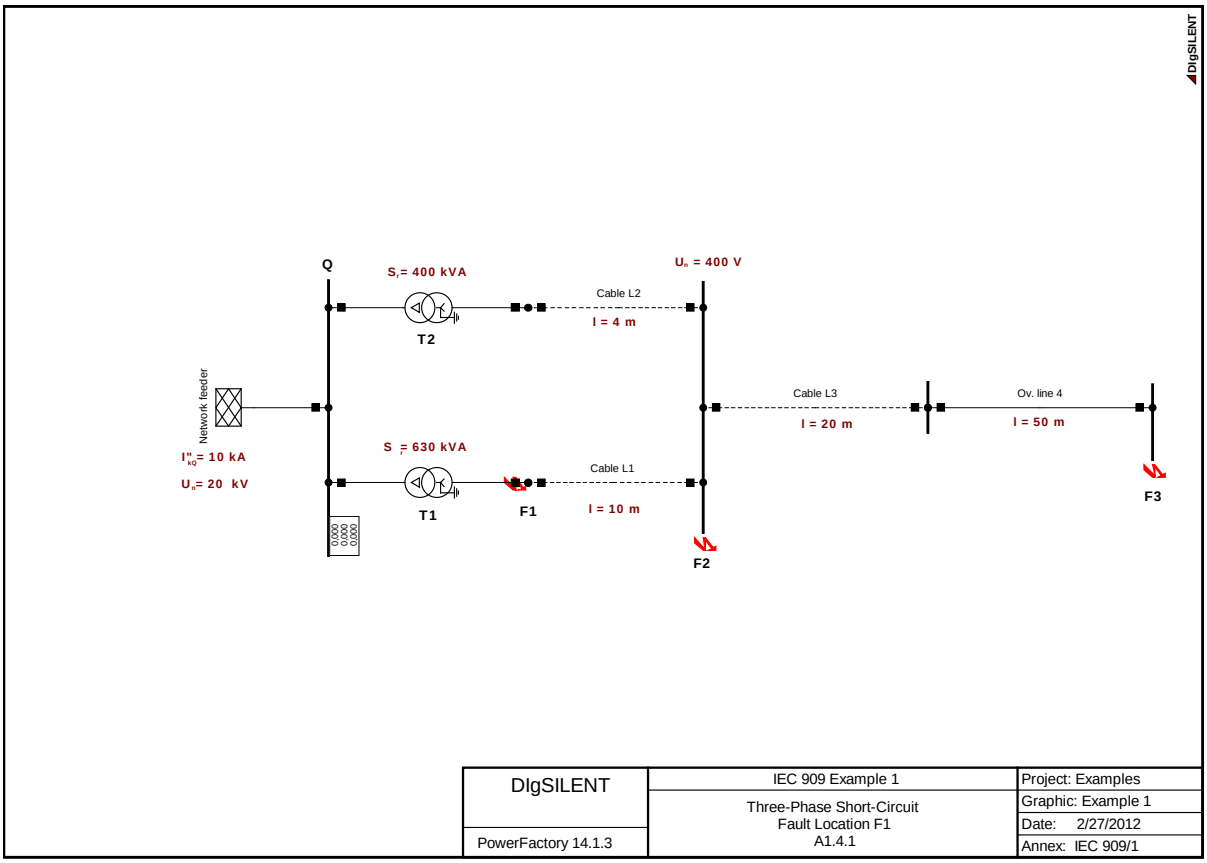


Figure 2.1 IEC Example 1: Low voltage network

2.2 Equipment Data

The data of the electrical equipment used in this low voltage example as defined in the TR 60909-4 is given in Table 1.

Table 1 Equipment data of the low voltage system: Example 1

Equipment	Data of Equipment
Network Feeder Q	U_{nQ}= 20 kV; I''_{kQ}= 10 kA R_Q= 0.1X_Q ,X_Q= 0.995Z_Q
Transformer T1 (Dyn5)	S_{rT}= 630 kVA; U_{rTHV}= 20kV; U_{rTLV}= 410 V u_{kr}= 4%; P_{krT}= 6,5 kW; $R_{(0)T}/R_T$= 1,0; $X_{(0)T}/X_T$= 0,95
Transformer T2 (Dyn5)	S_{rT}= 400 kVA; U_{rTHV}= 20kV; U_{rTLV}= 410 V u_{kr}= 4%; P_{krT}= 4,6 kW $R_{(0)T}/R_T$= 1,0; $X_{(0)T}/X_T$= 0,95
Cable L1	Two parallel four-core cables length= 10 m; 4x240 mm^2 Cu R= 0,077 Ω/km; X= 0,079 Ω/km $R_{(0)}$= 3,7R; $X_{(0)}$= 1,81X
Cable L2	Two parallel three-core cables length= 4 m; 3x185 mm^2 Al R= 0,208 Ω/km, X= 0,068 Ω/km $R_{(0)}$= 4,23R; $X_{(0)}$= 1,21X
Cable L3	Four-core cable length= 20 m; 4x70 mm^2 Cu R= 0,271 Ω/km; X= 0,087 Ω/km $R_{(0)}$= 3R; $X_{(0)}$= 4,461X
Overhead Line L4	length= 50 m; q_n= 50 mm^2 Cu; d= 0,4m R= 0,3704 Ω/km, X= 0,297 Ω/km $R_{(0)}$= 2R, $X_{(0)}$= 3X

For a detailed procedure on the calculation of the positive and zero sequence impedances, please refer to sections 3.2 and 3.3 of [1].

2.3 Entry of Equipment Data in *PowerFactory*

The following tables summarize the equipment data in the input format required in *PowerFactory* . All values correspond to the data in Table 1, IEC Example 1.

Table 2 Network Feeder data

Equipment	Data entered in *PowerFactory*					
	Sk''max MVA	Ik''max kA	R/X max.	Z2/Z1 max.	X0/X1 max.	R0/X0 max.
Network Feeder Q	346,41	10	0,1	1	1	0,1

Table 3 Transformer data

Equipment	Data entered in *PowerFactory*									
	Rtd. Pow. MVA	Nom Freq Hz	HV-rtd.Volt. kV	LV-Rtd.Volt. kV	uk %	Cop.Los. kW	Re(Shc Volt.) %	Ratio X/R	uk0 %	Shc Res.0 %
Transformer T1	0,4	50	20	0,41	4	4,6	1,15	3,331	3,889	1,15
Transformer T2	0,63	50	20	0,41	4	6,5	1,031	3,745	3,813	1,031

Table 4 Line data

Equipment	Data entered in *PowerFactory*					
	Parallel Cables	Length m	R' (20°C) Ohms/km	X' Ohms/km	R0 Ohms/km	X0 Ohms/km
Cable L1	2	10	0,208	0,068	0,87984	0,08228
Cable L2	2	4	0,077	0,079	0,2849	0,14299
Cable L3	1	20	0,271	0,087	0,813	0,38802
Overhead Line L4	1	50	0,3704	0,297	0,7408	0,891

2.4 IEC Short Circuit Command Options in *PowerFactory*

The short circuit calculation is launched by clicking on the "Calculate Short Circuit" button as shown in Figure 2.2.

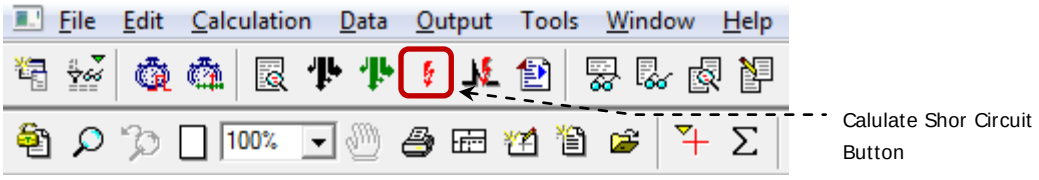


Figure 2.2 Location of the "Calculate Short Circuit" button in the main toolbar

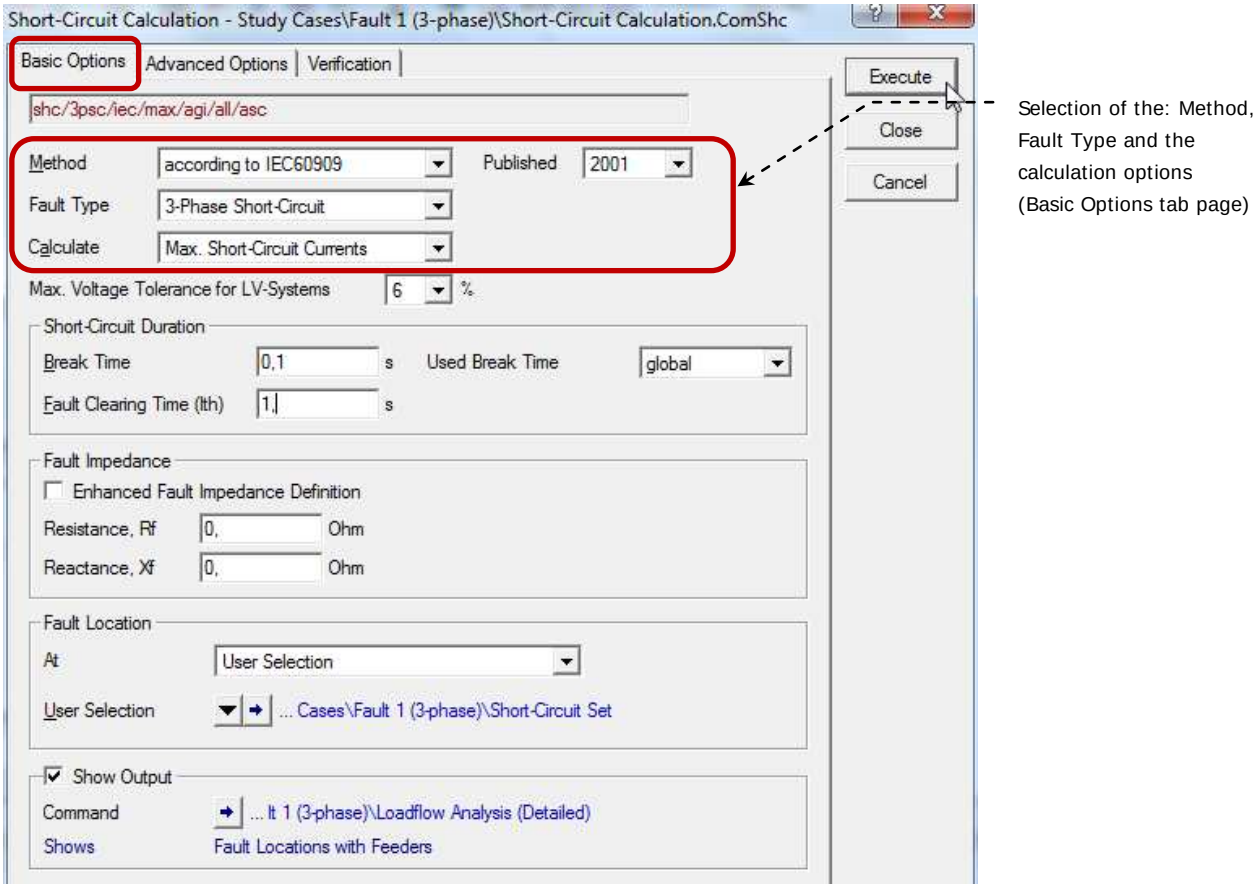
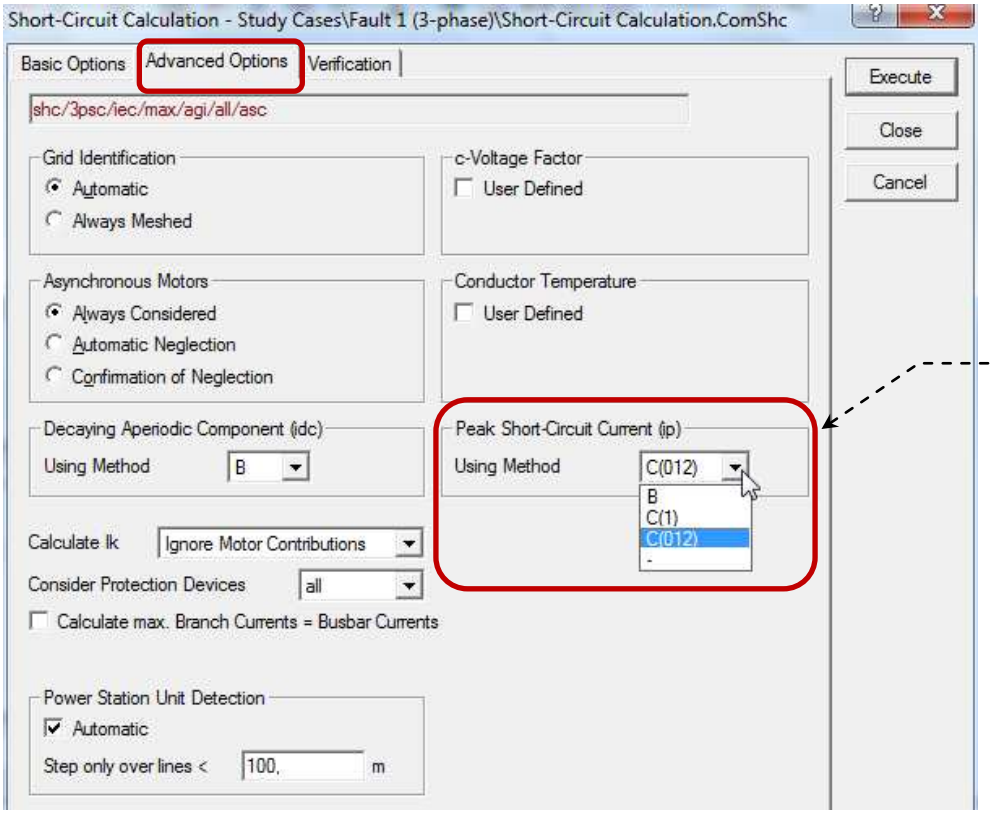


Figure 2.3 Basic options tab page of the short circuit command



Selection of the: Peak short circuit current method (Advanced Options tab page)

Figure 2.4 Advanced options tab page of the short circuit command

2.5 Comparison of Results

The following tables compare the results obtained with DIgSILENT *PowerFactory* [2] and the hand calculations given in [1].

2.5.1 Three-Phase Short-Circuit

Table 5 Comparison of 3 phase short circuit currents

Short Circuit Location	Results acc. to IEC-60909-4 (kA)		*PowerFactory* Results (kA)	
	I''k	$i_{p(c)}$	I''k	$i_{p(c)}$ [1]
F1	34,62	70,85	34,62	70,86
F2	34,12	69,10	34,12	69,07
F3	6,95	10,38	6,94	10,36

1 Calculated using method C(1)

2.5.2 Single-Phase Short-Circuit to Earth

Table 6 Comparison of single phase short circuit currents

Short Circuit Location	Results acc. to IEC-60909-4 (kA)		*PowerFactory* Results (kA)	
	I''k1	$i_{p1(c)}$	I''k1	$i_{p1(c)}$ [1]
F1	35,64	72,93	35,651	72,963
F2	34,98	70,84	34,912	70,679
F3	4,83	7,21	4,831	7,211

1 Calculated using method C(1)

The above results show a maximum difference of 0.2%. These differences can be further explained as a result of a three-figure number after the decimal comma used in the hand calculations in [1].

[2] Version 14.1.3

3 Example 4: IEC High Voltage Test Network

The following example is taken from the IEC Technical Report TR 60909-4[1], clause 6, page 119. In this example, a high voltage a.c. test network 380kV/110 kV/ 30 kV/10 kV, 50 Hz is given; in which three phase and line to earth short circuit currents are calculated at different locations.

3.1 System Description

The high voltage test network is illustrated in Figure 3.1. The system depicts two network feeders connected to a 110 kV system. Network feeder Q1 is connected to the 110 kV system via two network transformers (T3 and T4), while network feeder Q2 is considered directly connected to 110 kV. Additionally, several generators and asynchronous motors are considered.

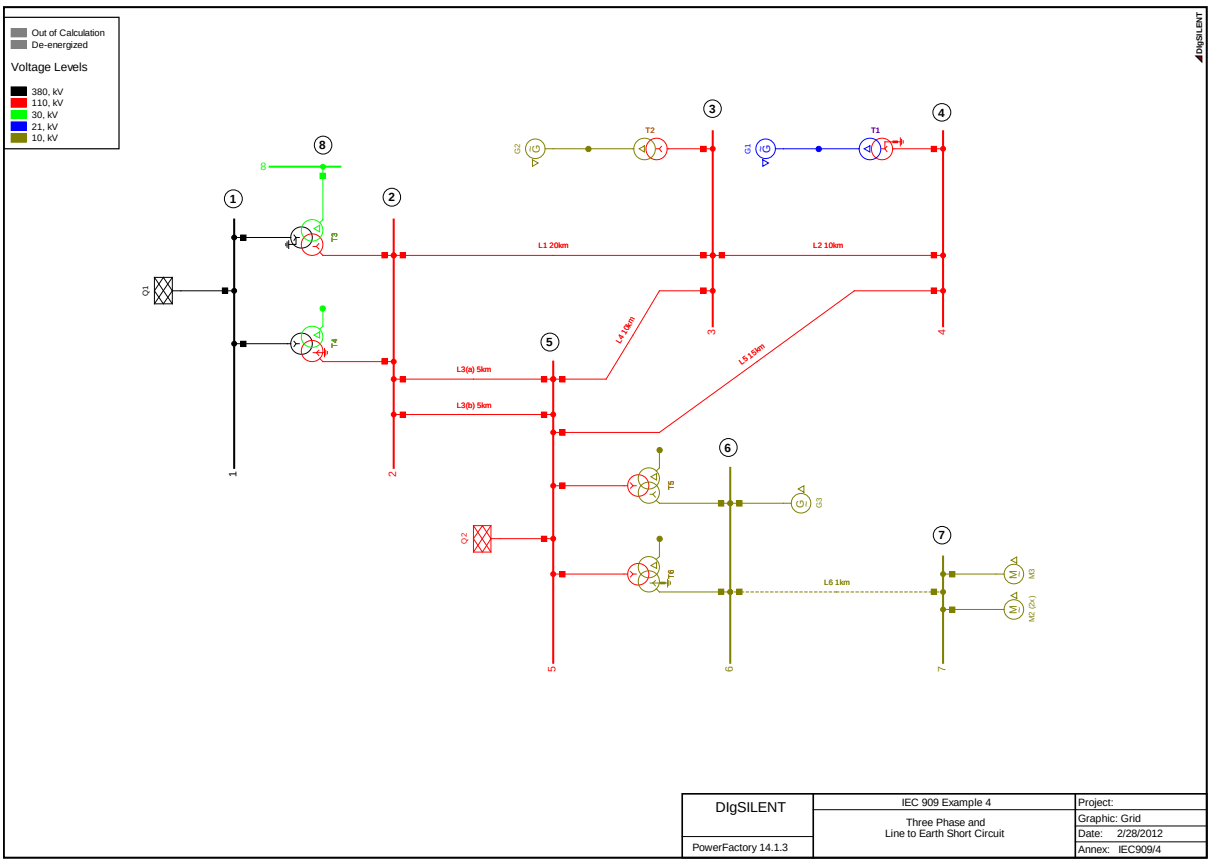


Figure 3.1 IEC Example 4: High voltage test network

3.2 Equipment Data

The data of the electrical equipment for the high voltage test system as defined in the TR 60909-4 is given in Table 7.

Table 7 Equipment data of the high voltage test network

Equipment	Data of Equipment
Network Feeder Q1	U_{nQ} = 380 kV; I''k_{Qmax} = 38 kA R_Q/X_Q = 0,1; $X_{(0)Q}/X_Q$ = 3; $R_{(0)Q}/X_{(0)Q}$ = 0,15
Network Feeder Q2	U_{nQ} = 110 kV; I''k_{Qmax} = 16 kA R_Q/X_Q = 0,1; $X_{(0)Q}/X_Q$ = 3,3; $R_{(0)Q}/X_{(0)Q}$ = 0,2
Power Station Unit S1 (G1+ T1)	U_{rG} = 21 kV; S_{rG} = 150 MVA; X''$_d$ = 0,14 p.u.; x_{dsat} = 1,8 p.u.; cosφ_{rG} = 0,85; R_G = 0,002 Ω (operated only in the overexcited region) U_{rTHV}/U_{rTLV} = 115 kV/21 kV; S_{rT} = 150 MVA; u_{kr} = 16 %; u_{Rr} = 0,5 %; YNd5 with onload tap-changer p_T = ±12 %; $X_{(0)T}/X_T$ = 0,95; $R_{(0)T}/R_T$ = 1,0
Power Station Unit S2 (G2+ T2)	U_{rG} = 10,5 kV; S_{rG} = 100 MVA; X''$_d$ = 0,16 p.u.; x_{dsat} = 2,0 p.u.; cosφ_{rG} = 0,9; R_G = 0,005 Ω (operated only in the overexcited region) p_G = ±7,5 % U_{rTHV}/U_{rTLV} = 120 kV/10,5 kV; S_{rT} = 100 MVA; u_{kr} = 12 %; u_{Rr} = 0,5 %; YNd5 without tap-changer $X_{(0)T}/X_T$ = 1,0; $R_{(0)T}/R_T$ = 1,0
Generator G3	U_{rG} = 10,5 kV; S_{rG} = 10 MVA; X''$_d$ = 0,1 p.u.; x_{dsat} = 1,8 p.u.; cosφ_{rG} = 0,8; R_G = 0,018 Ω p_G = ±5 % (for the calculation a constant value U_G = U_{rG} is assumed)
Network Transformers T3 and T4	Three-winding network transformers YNyn,d5 with on-load tap-changer at the high voltage side, p_T = ±16 % Star point earthing: T3 at the high-voltage side, T4 at the medium-voltage side. U_{rTHV} = 400 kV; U_{rTMV} = 120 kV; U_{rTLV} = 30 kV; S_{rTHV} = 350 MVA; S_{rTMV} = 350 MVA; S_{rTLV} = 50 MVA; u_{krHVMV} = 21 %; u_{krHVLV} = 10 %; u_{krMVLV} = 7 %; u_{RrHVMV} = 0,26 %; u_{RrHVLV} = 0,16 %; u_{RrMVLV} = 0,16 %; $X_{(0)TMV}/X_{TMVHV}$ = 2,1; $R_{(0)TMV}/R_{TMVHV}$ = 1,0
Network Transformers T5 and T6	Three-winding network transformer YNyn,d5, treated here as a two-winding transformer, i.e. u_{kr} = u_{krHVMV} U_{rTHV}/U_{rTMV} = 115 kV/ 10,5 kV; S_{rT} = 31,5 MVA; u_{kr} = 12 %; u_{Rr} = 0,5 %

Equipment	Data of Equipment
Asynchronous Motor M1	U_{rM} = 10 kV; P_{rM} = 5 MW; $\cos\varphi_{rM}$ = 0,88; η_{rM} = 97,5 %; I_{LR}/I_{rM} = 5; p = 1 (pair of poles)
Asynchronous Motor M2	Two parallel motors with P_{rM} = 2 MW each: U_{rM} = 10 kV; P_{rM} = 2 MW; $\cos\varphi_{rM}$ = 0,89; η_{rM} = 96,8 %; I_{LR}/I_{rM} = 5,2; p = 2
Reactor R1	X_{R1} = 22 W; $R_{R1} << X_{R1}$ (short-circuit limiting reactor)
Reactor R6	Arc-suppression coil for the 10 kV network with resonance neutral earthing
Overhead Line L1	Length = 20 km, Z1 = 0,12 + j 0,39 Ohms/km Z0= 0,32 + j 1,26 Ohms/km
Overhead Line L2	Length = 10 km, Z1 = 0,12 + j 0,39 Ohms/km Z0= 0,32 + j 1,26 Ohms/km
Overhead Line L3 (a)	Length = 5 km, Z1 = 0,12 + j 0,39 Ohms/km Z0= 0,52 + j 1,86 Ohms/km
Overhead Line L3 (b)	Length = 5 km, Z1 = 0,12 + j 0,39 Ohms/km Z0= 0,52 + j 1,86 Ohms/km
Overhead Line L4	Length = 10 km, Z1 = 0,096 + j 0,388 Ohms/km Z0= 0,22 + j 1,1 Ohms/km
Overhead Line L5	Length = 15 km, Z1 = 0,12 + j 0,386 Ohms/km Z0= 0,22 + j 1,1 Ohms/km
Cable L6	Length = 1 km, Z1 = 0,082 + j 0,086 Ohms/km

For a summary on the short circuit impedances of the electrical equipment in the positive sequence and the zero sequence systems in accordance with IEC 60909-0 including the correction factors K_G, K_S and K_T, please refer to section 6.2.2 of [1].

3.3 Entry of Equipment Data in *PowerFactory*

The following tables summarize the equipment data in the input format required by *PowerFactory* .All values are derived from those in Table 7and relate to IEC Example 4.

Table 8 Network Feeder data

Equipment	Data entered in *PowerFactory*					
	Sk''max MVA	Ik''max kA	R/X max.	Z2/Z1 max.	X0/X1 max.	R0/X0 max.
Network Feeder Q1	25010,81	38	0,1	1	3	0,1
Network Feeder Q2	3048,409	16	0,1	1	3,3	0,2

Table 9 2-Winding Transformer data

Equipment	Data entered in *PowerFactory*									
	Rtd. Pow. MVA	Nom Freq Hz	HV-rtd.Volt. kV	LV-Rtd.Volt. kV	uk %	Cop.Los. kW	Re(Shc Volt.) %	Ratio X/R	uk0 %	Shc Res.0 %
Transformer T1	150,	50,	115,	21,	16,	750,	0,5	31,984	15,2	0,5
Transformer T2	100,	50,	120,	10,5	12,	500,	0,5	23,979	12,	0,5

Table 10 3-Winding Transformer data

Equipment	Data entered in *PowerFactory*									
	HV-rtd.Pow. MVA	MV-rtd.Pow. MVA	LV-rtd.Pow. MVA	HV-rtd.Volt. kV	MV-rtd.Volt. kV	LV-rtd.Volt. kV	HV-MV-Shc.Volt. %	MV-LV-Shc.Volt. %	LV-HV Shc.Volt. %	HV-MV Cop.Los. kW
Transformer T3 and T4	350,	350,	50,	400,	120,	30,	21	7	10	910
Transformer T5 and T6	31,5	31,5	31,5	115,	10,5	10,5	12	12	12	157,5

Table 11 Generator data

Equipment	Data entered in *PowerFactory*					
	App.Pow. MVA	Nom.Volt. kV	Pow.Fact.	xd''sat	rstr	xdsat
Generator G1	150	21	0,85	0,14	0,00068027	1,8
Generator G2	100	10,5	0,9	0,16	0,00453515	2,0
Generator G3	10	10,5	0,8	0,1	0,00163265	1,8

Table 12 Asynchronous Motor data

Equipment	Data entered in *PowerFactory*					
	Rtd. MechPow. kW	Nom.Volt. kV	Pow.Fact.	Efficiency %	Pair of poles	I_{LR}/I_{rM}
Motor M1	5000	10	0,88	97,5	1	5,0
Motor M2	2000	10	0,89	96,8	2	5,2

Table 13 Line data

Equipment	Data entered in *PowerFactory*					
	Parallel Cables	Length km	R' (20°C) Ohms/km	X' Ohms/km	R0 Ohms/km	X0 Ohms/km
Overhead Line L1	1	20,	0,12	0,39	0,32	1,26
Overhead Line L2	1	10,	0,12	0,39	0,32	1,26
Overhead Line L3(a)	1	5,	0,12	0,39	0,52	1,86
Overhead Line L3(b)	1	5,	0,12	0,39	0,52	1,86
Overhead Line L4	1	10,	0,096	0,388	0,22	1,1
Overhead Line L5	1	15,	0,12	0,386	0,22	1,1
Overhead Line L6	1	1,	0,082	0,086	0,082	0,086

3.4 IEC Short Circuit Command Options in *PowerFactory*

The short circuit calculation is launched by clicking on the "Calculate Short Circuit" button as shown in Figure 3.2.

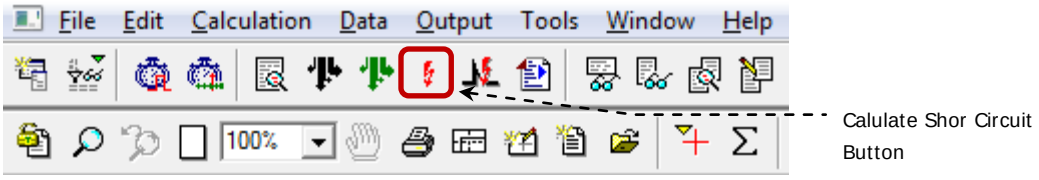


Figure 3.2 Location of the "Calculate Short Circuit" button in the main toolbar

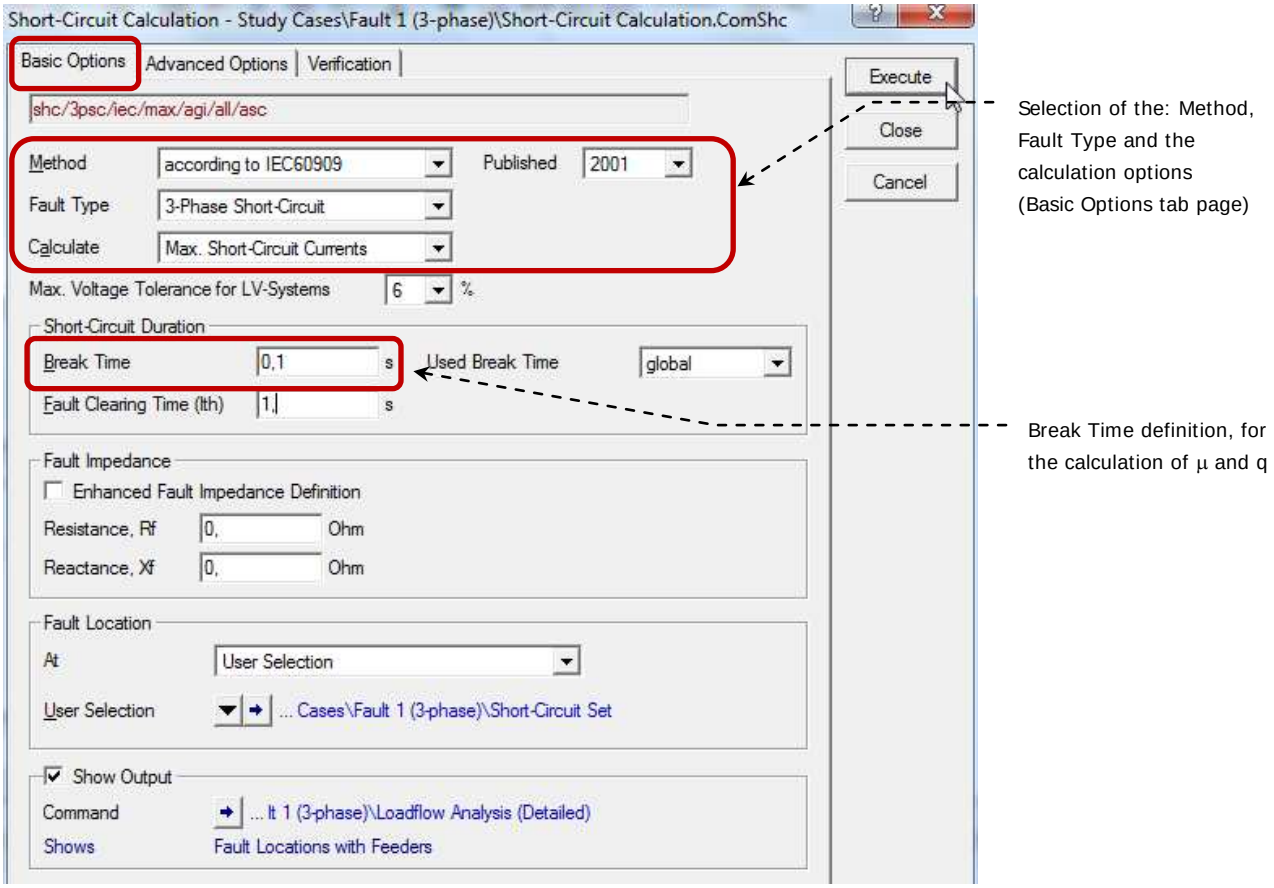


Figure 3.3 Basic options tab page of the short circuit command

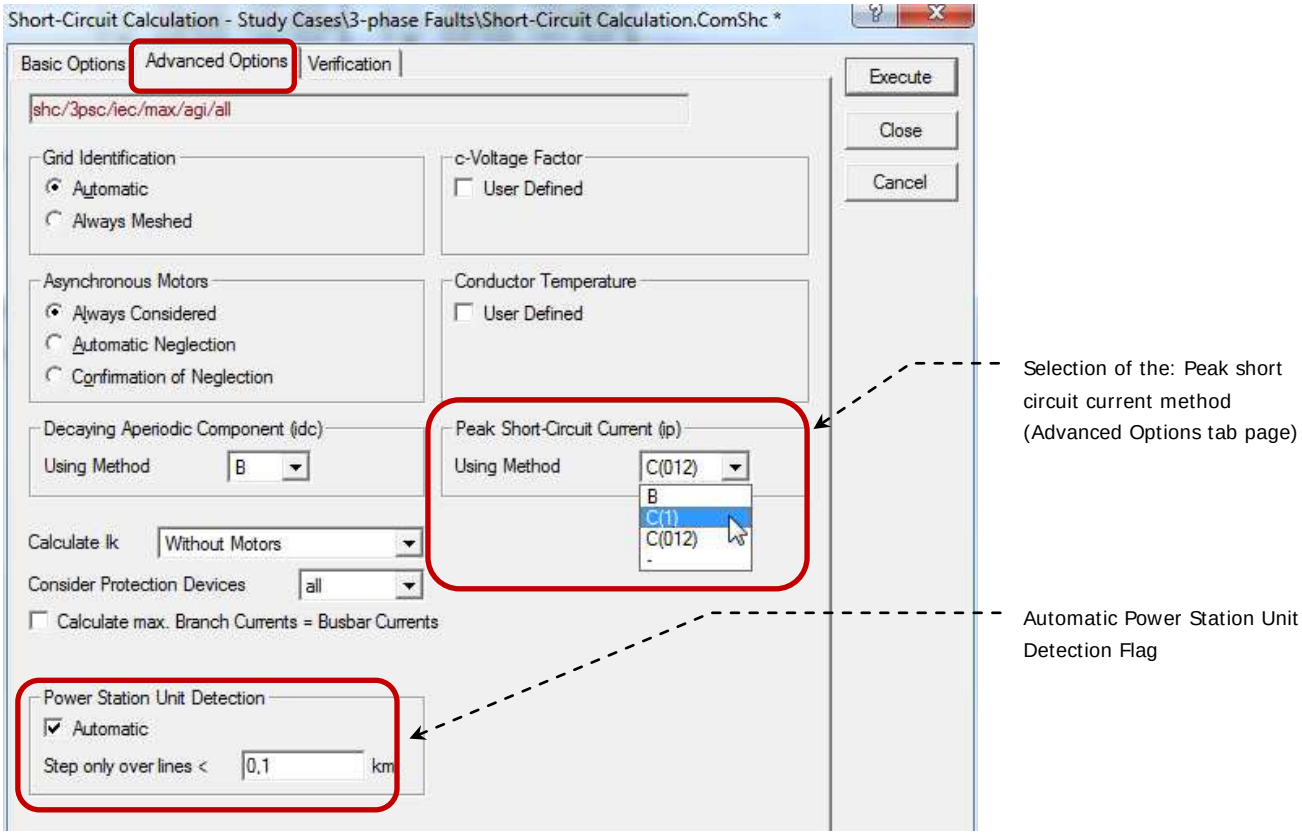


Figure 3.4 Advanced options tab page of the short circuit command

3.5 Comparison of Results

The following tables compare the results obtained with PowerFactory and the calculations obtained in [1].

3.5.1 Three phase short-circuit

Table 14 Results acc. to IEC-60909-4 (3 phase short circuit)

Short Circuit Location	U_n kV	I''_{k1} kA	$i_p(20)$ kA	i_b kA	i_k kA
1	380	40,6447	100,5677	40,645	40,635
2	110	31,7831	80,6079	31,570	31,663
3	110	19,6730	45,8111	19,388	19,623
4	110	16,2277	36,8427	16,017	16,196
5	110	33,1894	83,4033	32,795	32,997
6	10	37,5629	98,1434	34,028	34,356
7	10	25,5895	51,6899	23,212	22,276
8	30	13,5778	36,9227	13,578	13,573

Table 15 Results obtained with PowerFactory (3 phase short circuit)

Short Circuit Location	U_n kV	I''_{k1} kA	$i_p(20)$ [1)] kA	i_b kA	i_k kA
1	380	40,6447	100,5677	40,642	40,635
2	110	31,7831	80,6080	31,569	31,663
3	110	19,6730	45,8111	19,388	19,623
4	110	16,2277	36,8427	16,017	16,196
5	110	33,1894	83,4033	32,795	32,997
6	10	37,5629	98,1434	34,028	34,356
7	10	25,5894	51,6900	23,212	22,276
8	30	13,5778	36,9227	13,578	13,573

1) Peak Short circuit current calculated using method C(1)

3.5.2 Single phase short-circuit to earth

Table 16 Results acc. to IEC-60909-4 (Line to earth short circuit currents)

Short Circuit Location	U_n kV	I''_{k1} kA	$i_p(1)$ kA	$i_p(012)$ kA
2	110	15,9722	40,5086	39,9641
3	110	10,4106	24,2424	24,2635
4	110	9,0498	20,5463	21,0415
5	110	17,0452	42,8337	41,4303

Table 17 Results obtained with PowerFactory (Line to earth short circuit currents)

Short Circuit Location	U_n kV	I''_{k1} kA	$i_p(1)^{1)}$ kA	$i_p(012)^{2)}$ kA
2	110	15,9722	40,5086	39,9641
3	110	10,4106	24,2424	24,2635
4	110	9,0498	20,5464	21,0415
5	110	17,0452	42,8337	41,4303

1) Peak Short circuit current calculated using method C(1)

1) Peak Short circuit current calculated using method C(012)

4 References

[1] IEC Technical Report 60909-4, "Short-circuit currents in three-phase a.c. systems- Part 4: Examples for the calculations of short-circuit currents". First Edition, 2000-7.

[2] IEC International Standard 60909-0, "Short-circuit currents in three- phase a.c. systems- Part 0: Calculation of currents". First Edition, 2001-7.